

Replication Study
On
‘The Miracle of Microfinance? Evidence from a Randomized Evaluation’.

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Abstract

This study replicates and extends the randomized evaluation of group-lending microfinance conducted by Banerjee et al. (2015) in Hyderabad, India. Using the original experimental design and household survey data, the replication confirms that expanded access to microfinance produces modest and heterogeneous short-run effects including increases in business investment and asset ownership among households with pre-existing entrepreneurial activity, alongside limited changes in business profits, consumption, and labor supply. Building on these results, the analysis examines the temporal evolution of treatment effects across 2 surveys. The evidence indicates that asset gains for business-owning households persist, while effects on investment, profits, revenues, and labor supply remain small and statistically weak across periods. A complementary examination of borrowing behavior, based on within-household changes and instrumental-variables estimation, shows no strong evidence of substitution from formal to informal borrowing but does reveal shifts in microfinance borrowing consistent with program exposure. Overall, the findings align with a broader literature that characterizes microfinance impacts as modest, selective, and highly dependent on baseline business characteristics and local credit-market conditions.

Introduction

Microfinance has long been regarded as a promising instrument for poverty alleviation and financial inclusion. The central idea is that access to small loans without collateral requirements can support entrepreneurial activity, enable the accumulation of productive assets, and stabilize consumption among households that lack access to formal banking systems. Evidence from major randomized controlled trials conducted over several decades, however, has produced a more complex and often more modest assessment of the economic and social effects associated with microfinance programs.

A prominent contribution to this body of research is the study by Banerjee, Duflo, Glennerster, and Kinnan published in 2015 under the title “The Miracle of Microfinance Evidence from a Randomized Evaluation.” This study provides the first large scale randomized evaluation of a group lending microfinance program by taking advantage of the phased expansion of Spandana in Hyderabad, India. The research design assigned 104 urban neighborhoods to treatment or control status and followed households across two detailed survey rounds. This approach made it possible to estimate the causal effects of access to microfinance on business investment, business profits, consumption, borrowing behavior, and labor supply. The main results show moderate increases in business investment and asset ownership among households with existing enterprises, along with limited evidence of improvements in broader outcomes such as consumption, health, and education. Substantial variation across households is also observed, with gains concentrated among those already engaged in business activity rather than among lower income households.

This replication study reproduces the central ordinary least squares estimates from the original analysis using the same dataset to evaluate the effects of microfinance expansion on household and business outcomes. Two extensions complement this replication. The first extension examines the persistence of treatment effects by comparing outcomes between the first endline survey and the second endline survey to determine whether microfinance effects increase, diminish, or remain stable with continued exposure. The second extension focuses on borrowing behavior by studying changes in the use of formal and informal credit sources between survey rounds. This investigation relies on changes observed within households as well as instrumental variables approach that uses randomized program assignment to identify the causal effect of microfinance take up on credit portfolios.

These extensions address ongoing discussions in the development economics literature concerning the durability of microfinance impacts and the extent to which access to microfinance alters the structure of household borrowing. The resulting evidence supports the modest and uneven effects reported in earlier work while providing additional insight into household level adjustments over time. Patterns observed in the data indicate persistent increases in asset accumulation for some households, as well as a shift toward greater reliance on informal credit and reduced participation in formal credit markets after access to microfinance. These findings deepen the understanding of how microfinance influences economic decisions and offer guidance for policy debates regarding the design and targeting of financial inclusion programs intended to serve low-income populations.

Theoretical Framework

Microfinance programs are founded on the idea that households with limited income and small-scale entrepreneurs in developing countries often encounter significant obstacles when attempting to obtain formal credit, largely because acceptable collateral is unavailable and traditional financial institutions classify these borrowers as high risk. The provision of group loans without collateral is intended to relax these financial constraints and allow recipients to invest in productive assets, expand business operations, and manage consumption and risk more effectively. Standard economic reasoning predicts that access to microfinance should increase investment, strengthen business asset holdings, raise income, and possibly influence labor supply when loan funds are directed toward productive activities.

However, outcomes are not uniform across all households, since some recipients use loan funds for short term consumption needs or experience limited success with new business attempts, leading to improvements that are modest or temporary. Access to microfinance can also alter engagement with informal credit markets, although the direction and intensity of this change depend on the relative benefits offered by formal and informal lending arrangements in each setting. These mechanisms provide the conceptual foundation for the analysis of short term and longer-term effects of microfinance, as well as for the investigation of shifts in borrowing patterns across credit markets.

Literature Review

Since Banerjee, Duflo, Glennerster, and Kinnan (2015) presented one of the earliest randomized evaluations of microfinance, a substantial research literature has reexamined both the strengths and the limitations of microcredit programs. Initial enthusiasm regarding the potential of microcredit to lift households out of poverty has gradually shifted toward a more cautious view, informed by evidence of considerable variation in program effects across settings. Numerous studies show increases in business investment and business profits among households already engaged in entrepreneurial activity, while effects on consumption, education, health, and other broad welfare indicators tend to be limited for the general population.

A significant development in this assessment has been the growing use of meta analytic and hierarchical methods that combine results from many independent evaluations. Chliova, Brinckmann, and Rosenbusch (2015) assemble findings from 90 studies and report that microcredit produces positive but moderate effects across economic and noneconomic outcomes. Meager (2022) uses a Bayesian hierarchical framework to analyze data from seven randomized controlled trials and finds that overall treatment effects are small and highly dispersed, with the largest gains concentrated among households that already operate businesses. These contributions emphasize the importance of understanding heterogeneity across different types of households and baseline conditions, a theme that plays a central role in continued empirical work on microfinance.

More recent research has shifted attention toward macro level consequences of microfinance expansion. Jensen and Naveed (2025) study panel data from 123 countries covering the period from 2005 to 2019 and document that increased microfinance activity is associated with reductions in national income inequality, including declines in the Gini coefficient and poverty headcounts, along with increases in income shares for the lowest quintile. These associations remain after accounting for national differences in education, trade, and institutional quality. Biru (2024) also presents evidence from a meta analysis of financial inclusion programs and finds that access to credit and savings instruments improves household welfare, although the magnitude of improvement is generally modest rather than transformative. When viewed together, these studies suggest that microfinance can generate incremental gains in economic well being and equity, even though the size and distribution of these gains vary substantially across contexts.

The broader literature following Banerjee and coauthors therefore characterizes microfinance as a tool that delivers moderate but uneven effects shaped by local conditions and borrower characteristics. This body of evidence underscores the value of analyzing outcomes separately by household type, by business status, and across time, in order to understand the channels through which microfinance produces sustained or temporary changes in economic behavior.

Early randomized controlled trials consistently reported modest average treatment effects from microcredit programs. These effects included increases in business investment and business assets, but little systematic change in household consumption, business profits, or indicators of empowerment. The question of persistence has emerged as an important theme in subsequent research, given the possibility that microfinance may support long term poverty reduction only if initial gains endure or expand over time. Evidence showing the strongest persistence in business asset accumulation aligns with economic theories that highlight the role of capital indivisibilities and significant adjustment costs in small enterprises, as described in work by de Mel, McKenzie, and Woodruff (2008). At the same time, limited or declining effects on business profits and business investment point to a central puzzle. In many environments, increases in asset ownership do not produce sustained increases in income or improvements in overall welfare. This pattern suggests that microfinance may support a single episode of capital acquisition without creating sustained business growth or continuous entrepreneurial development. For many households, access to credit enables the initial purchase of durable inputs but does not necessarily provide the conditions required for long term or transformative economic progress.

Data and Descriptive Statistics

This replication paper uses the publicly available replication dataset from “*The Miracle of Microfinance? Evidence from a Randomized Evaluation*” by Banerjee, Duflo, Glennerster, and Kinnan (2015). The dataset was accessed through the American Economic Association RCT Replication Repository, which provides the full set of household survey files used in the original study. The data cover 104 slum neighborhoods in Hyderabad, India, where access to Spandana microfinance was allocated by random assignment. Households were interviewed at baseline and in two endline rounds, producing extensive information on business activity, borrowing patterns,

consumption, assets, household structure, and labor supply. All variables used in the replication and extensions originate from these public survey files.

The main explanatory variable is the randomized treatment assignment at the neighborhood level. Areas selected for early access to microfinance are coded as treated, and areas not yet selected form the comparison group. This assignment indicator serves as the treatment measure for the replication results and for both extensions.

Outcome variables follow the structure of the original study. Business outcomes include monthly profits, business revenues, investment during the past year, the value of business assets, the number of business employees, and hours worked in household enterprises. Additional outcomes describe total labor hours, nondurable and durable expenditure, food expenditure, total monthly expenditure, and several borrowing indicators that distinguish between formal credit, informal credit, and microfinance institution loans. For the second endline survey, variable names in the dataset include a final numeral, and these variables are retained without alteration aside from renaming for clarity.

Each regression includes the baseline area level controls used in the original paper. These controls are neighborhood population, total outstanding household debt, the number of businesses recorded at baseline, average per capita expenditure, and the baseline literacy rate. This set of controls appears in every specification, whether in the direct replication or in the extended analyses. Survey weights and clustering at the neighborhood level are also applied to maintain consistency with the original design.

The extended analyses require additional outcome constructions. The persistence analysis estimates treatment effects separately for the first and second endline survey. This approach allows a comparison of estimates across two time periods and reveals whether treatment effects increase, diminish, or remain stable as exposure to microfinance continues. The borrowing behavior extension examines transitions in credit use. For this analysis, differences between the first and second endline survey are calculated for each borrowing outcome, creating within household changes in formal and informal loan use. For the instrumental variables specification, actual borrowing from microfinance institutions in the second endline survey is treated as an endogenous variable, and randomized treatment assignment serves as the instrument. This approach isolates the effect of borrowing take up from the effect of simple eligibility.

Observations are excluded only when missing in the original survey files and that makes the differences in sample sizes in survey structure. Business outcomes appear only for households reporting business activity, and several borrowing and expenditure questions include skipping patterns. No additional restrictions or trimming procedures are introduced.

Descriptive statistics provide an overview of the data used throughout the analysis. The table presents baseline characteristics for households in treatment and control neighborhoods, covering demographic composition, borrowing activity, business outcomes, and consumption measures. The table includes observations, means, standard deviations, and treatment control differences with corresponding p values.

Table 1A: Baseline Summary statistics

Variables	Obs	Control Mean	Control SD	Coeff	p-value
<i>Household composition</i>					
Number members	1220	5.038	1.666	0.095	0.303
Adults	1220	3.439	1.466	-0.011	0.873
Children	1220	1.599	1.228	0.104	0.098
Male head	1216	0.907	0.290	-0.012	0.381
Head's age	1216	41.150	10.839	-0.243	0.676
Head with no education	1216	0.370	0.483	-0.008	0.787
<i>Access to credit</i>					
Loan from Spandana	1213	0.000	0.000	0.007	0.195
Loan from other MFI	1213	0.011	0.103	0.007	0.453
Loan from a bank	1213	0.036	0.187	0.001	0.859
Informal loan	1213	0.632	0.482	0.002	0.958
<i>Amount borrowed from</i>					
Any type of loan	1213	0.680	0.467	0.002	0.942
Spandana	1213	0.000	0.000	68.966	0.192
Other MFI	1213	201.154	2742.364	170.357	0.568
Bank	1213	7438.170	173268.340	-5419.697	0.279
Informal loan	1213	28460.016	65312.156	-570.460	0.856
Total	1213	37892.004	191291.560	-5878.555	0.343
<i>Self-employment activities</i>					
Number of activities	1220	0.320	0.682	-0.019	0.579
Activities managed by women	1220	0.145	0.400	-0.007	0.750
Share of HH managed by women	295	0.488	0.482	-0.006	0.904

<i>Businesses</i>					
Revenue/month (Rs)	295	15990.644	53488.746	4501.229	0.539
Expense/month (Rs)	295	3617.416	26144.037	640.552	0.751
Investment/month (Rs)	295	384.712	3156.815	14.478	0.959
Employment(employees)	295	0.169	0.828	0.255	0.148
Self-employment(hours/week)	295	76.315	66.054	-4.587	0.414
<i>Businesses (All household)</i>					
Revenue/month (Rs)	1220	3866.590	27146.787	903.649	0.626
Expense/month (Rs)	1220	874.703	12932.557	116.496	0.812
Investment/month (Rs)	1220	93.025	1559.052	-0.098	0.999
Employment(employees)	1220	0.041	0.413	0.057	0.166
Self-employment(hours/week)	1220	18.453	46.054	-1.801	0.400
<i>Consumption (per household per week)</i>					
Total consumption (Rs)	1220	4888.431	4074.372	269.683	0.232
Nondurables consumption (Rs)	1220	4734.686	3839.803	251.819	0.235
Durables Consumption (Rs)	1220	153.746	584.594	17.863	0.531
Asset Index	1220	1.941	0.829	0.027	0.669

Empirical Framework

Original Paper: Experimental Design and Analytic Approach

Banerjee et al. (2015) conducted a randomized controlled trial to evaluate the impact of introducing group-lending microfinance (Spandana) in slum neighborhoods of Hyderabad, India. The randomization assigned 104 urban neighborhoods to either treatment or control, yielding experimental variation in household access to microfinance. Treatment status was randomized at the area level, creating clear separation between intervention and comparison groups.

Households were surveyed at baseline and again at two endline rounds. The public-use dataset tracks business activity, financial outcomes, consumption, borrowing, and labor supply. Outcomes are measured both immediately after program introduction (Endline 1) and several years later (Endline 2).

The main estimation strategy is intent-to-treat analysis via OLS regression, applied separately for each outcome and survey wave:

$$Y_{ia} = \alpha + \beta \cdot \text{Treatment}_a + X'_a\gamma + \varepsilon_{ia}$$

Where:

- Y_{ia} is the outcome of interest for household i in area a , such as business profit (last 30 days), business investment (last 12 months), assets, revenue, or total labor hours per week.
- Treatment_a is a binary indicator of randomized eligibility for microfinance.
- X_a includes area-level control: population, total outstanding debt, number of businesses, mean per-capita expenditures, and literacy rates which were measured at baseline.
- ε_{ia} is the error term.

Survey weights for each wave are incorporated to ensure appropriate population representation. Standard errors are clustered at the area level, the unit of randomization, addressing intra-area correlation in outcomes.

This experimental design and estimation framework leverages random assignment for credible causal inference, enabling direct measurement of the average effect of microfinance access on targeted business and labor outcomes.

Extensions: Persistence and Borrowing Behavior Dynamics

Building on the core replication, the analysis is extended in two directions motivated by key policy and empirical questions.

1. Persistence of Treatment Effects

To evaluate whether the effects of microfinance endure, fade, or change over time, outcomes are regressed on treatment status separately at Endline 1 and Endline 2. Independent coefficients β_1 and β_2 are estimated for each period:

$$Y_{ia,t} = \alpha_t + \beta_t \cdot \text{Treatment}_a + X'_a \gamma_t + \varepsilon_{ia,t}$$

where t indexes the survey round.

This analysis uses the same controls, weights, and clustering as in the core replication, facilitating direct comparison of treatment impacts across both survey waves for investment, asset accumulation, business profits, revenues, and labor supply.

2. Evolution of Borrowing Behavior

To assess whether microfinance alters household borrowing portfolios, the following two empirical strategies are implemented:

A. Difference-in-Differences OLS:

Within-household changes in borrowing outcomes are examined using first-difference regressions:

$$\Delta Y_{ia} = \alpha + \beta \cdot \text{Treatment}_a + X'_a \gamma + \varepsilon_{ia}$$

where ΔY_{ia} is the change in formal or informal borrowing status (Endline 2 minus Endline 1), coded as binary indicators.

This specification absorbs household-level time-invariant factors, directly estimating whether assignment to microfinance drives transitions in formal or informal credit use over time.

B. Instrumental Variables (IV):

To estimate the causal effect of microfinance take-up, actual borrowing from microfinance institutions at Endline 2 ($\text{AnyMFI}_{ia,2}$) is instrumented using randomized treatment status:

$$Y_{ia,2} = \alpha + \theta \cdot \widehat{\text{AnyMFI}}_{ia,2} + X'_a \gamma + \varepsilon_{ia,2}$$

The IV approach yields the local average treatment effect (LATE), identifying the effect for compliers (households whose borrowing status is actually shifted by program access).

All extension models include the same set of baseline controls and employ area-level clustered standard errors and survey weights. No additional robustness or sensitivity checks were conducted; variable construction and sample restrictions precisely match established code and original analytic design.

Results

Persistence of Microfinance Effects Over Time

Table 2 presents the effect of microfinance treatment on business outcomes and labor supply at both survey rounds.

Table 2: Microfinance Effects Over Time

Outcome	Endline 1 (Coef., SE)	Endline 2 (Coef., SE)
Business Profits (Rs.)	354.3 (313.5)	542.0 (371.7)
Business Investment (Rs.)	390.9 (212.7)	-133.7 (206.6)
Business Assets (Rs.)	597.5 (383.5)	1260.8** (529.8)
Business Revenue (Rs.)	926.6 (1182.5)	266.1 (526.5)
Labor Supply (hours/week)	0.74 (2.25)	-1.24 (1.54)

Note: Double asterisk denotes significance at $p < 0.05$. Standard errors are in parentheses.

The most robust and policy-relevant finding is the statistically significant increase in business assets at Endline 2. Treatment area households owned, on average, 1261 Rupees more in productive assets than control households, roughly double the Endline 1 estimate. This effect is not only statistically significant but also substantively meaningful, demonstrating that repeated or continued access to microfinance allows some households to sustain and expand physical capital over time. This supports the economic theory that capital constraints limit micro-enterprise growth, and that relaxing these constraints through access to formal loans has persistent consequences for productive investment, particularly in asset-rich activities.

In contrast, the treatment effect on business profits, while positive at both rounds, remains statistically indistinguishable from zero. Confidence intervals are wide, including both negative and positive values, so there is little evidence that asset accumulation translated consistently into higher business output or net income for most recipients. Business investment has an initial positive and marginally significant effect on Endline 1 but becomes negative and insignificant at follow-up. This pattern may reflect investment that is concentrated at the beginning of the credit cycle. Loan funds are often deployed immediately after receipt, yet limited returns or exposure to business shocks can lead households to reduce activity or have trouble maintaining consistent profits. Revenue and labor supply show no meaningful changes, indicating that expanded access to finance does not transform overall economic participation or generate broad employment effects within this context.

The results highlight both the potential and the limits of microfinance. The increase in business assets indicates a durable improvement for a subset of entrepreneurs who face credit constraints. At the same time, the absence of measurable effects on business performance indicators or labor

outcomes shows that microfinance access does not ensure sustained or transformative economic growth across the full household population.

Evolution of Borrowing Behavior: Formal and Informal Credit

The second extension assesses whether microfinance enables households to shift away from informal lending and toward formal financial services.

Table 3: Changes in Borrowing Behavior

Model / Outcome	OLS Δ Formal (SE)	OLS Δ Informal (SE)	IV Formal (SE)	IV Informal (SE)
Treatment/Takeup	-0.094*** (0.022)	0.061** (0.020)	0.584 (5.548)	1.001 (17.433)
Constant	0.326* (0.126)	-0.131 (0.072)	0.203 (2.313)	0.492 (7.249)
Observations	6863	6100	6142	6142

Note: Triple and double asterisks denote significance at standard levels.

OLS estimates indicate substantial and statistically significant changes in household credit patterns following treatment. Assignment to a microfinance program area produced a 9.4 percentage point decline in formal borrowing and a 6.1 percentage point rise in informal borrowing relative to control areas. These magnitudes are considerable when compared with baseline borrowing rates and are estimated with precision. From an economic perspective, the pattern suggests that formal microfinance did not replace existing informal sources of credit. Instead, many households continued to rely on informal lenders or increased their reliance on such lenders after the introduction of microfinance. This behavior may reflect restrictive repayment structures, administrative procedures within microfinance institutions or banks that are poorly aligned with household financial needs, long standing social relationships with informal lenders, or the insurance and risk sharing features embedded in traditional informal arrangements that formal financial products cannot easily match.

IV estimates obtained through a two stage least squares framework, which use randomized treatment as an instrument for formal loan use, display considerable noise and fail to achieve statistical significance. The absence of clear IV effects indicates that program access alone may have heterogeneous or limited influence on borrowing transitions, or that the first stage relationship is weak, resulting in imprecise second stage estimates.

The overall pattern challenges a central justification often used to support microfinance expansion, namely the expectation that greater availability of formal credit will reduce reliance on high cost informal lenders and mitigate financial vulnerability. In the present context, microfinance expansion increased informal borrowing activity rather than displacing it, highlighting the need for deeper investigation into borrower preferences, loan product design, repayment rigidity, and structural features of the local credit environment.

Overall, the extended analysis demonstrates that access to microfinance supports persistent and economically meaningful accumulation of productive assets for a segment of households. At the same time, access does not ensure sustained improvements in business investment, profitability, or labor supply. Moreover, microfinance availability did not reduce informal borrowing but instead coincided with greater engagement with informal credit networks. These findings emphasize the importance of incorporating real world complexity into both policy formulation and theoretical work on microfinance, particularly in environments where formal and informal credit systems operate simultaneously and serve distinct economic and social functions.

Contribution and Policy Implications

The first major result concerns the sustained and expanding effect of microfinance on business asset accumulation. Households located in treatment areas recorded a statistically significant and sizable increase in productive assets at the second endline survey, reaching a level that is approximately twice as large as the effect measured at the first endline survey. This pattern aligns with theories emphasizing the importance of easing credit constraints and addressing fixed costs of investment faced by microenterprises. Microfinance appears capable of generating durable improvements in the stock of productive assets. At the same time, the lack of persistent gains in business profits, business revenue, or labor supply indicates limited translation of this capital growth into long term improvements in income or household welfare. The decline in business investment effects by the second endline survey further suggests that initial capital injections are not consistently sustained or expanded.

The second central result contributes directly to research on credit market behavior. The difference in differences estimates indicate that microfinance access is associated with a significant reduction in formal borrowing and a significant increase in informal borrowing between the first and second endline surveys. This pattern implies that microfinance products with rigid repayment obligations

may not replace informal credit but instead function alongside traditional lenders. The resulting rise in total household borrowing and increased reliance on informal credit sources raises concerns regarding financial vulnerability and may indicate a cycle in which informal loans are used to meet fixed repayment schedules imposed by microfinance institutions rather than serving as a pathway to financial inclusion.

Limitations and Future Research

The strength of empirical conclusions is influenced by several methodological constraints. The persistence analysis is based entirely on the intent to treat framework, which captures the effect of program access but may underestimate the true impact for households that actively borrow from microfinance institutions. The instrumental variables estimate used to identify the local average treatment effect of microfinance take up are highly imprecise, limiting the reliability of causal conclusions regarding borrowing transitions. In addition, reliance on area level baseline controls may not fully address unobserved and time constant household characteristics that shape both participation in microfinance programs and long-term economic outcomes.

Future work should advance along several lines. First, further analysis is needed to identify baseline household characteristics such as pre-existing entrepreneurial activity, asset ownership, or income level that may explain the persistence of asset accumulation. Second, research using high frequency panel data on financial flows, liquidity constraints, coping responses, and exposure to shocks would help clarify the mechanisms behind rising informal borrowing. Understanding whether this pattern is driven by risk sharing needs, repayment rigidity, or strategic borrowing behavior is essential for policy development. Finally, studies with longer time horizons are needed to determine whether continued asset accumulation can eventually generate transformative improvements in income or consumption that emerge only after a delayed period.

Conclusion

This study builds on the established findings of the randomized controlled trial conducted by Banerjee, Duflo, Glennerster, and Kinnan (2015) by using data from two endline survey waves to produce a detailed assessment of the time dependent effects of microfinance access. The extended analysis confirms the original pattern of moderate and uneven short run impacts and reveals an

important long-term structure in household outcomes. Evidence indicates a combination of persistent capital accumulation for a subset of households alongside a significant restructuring of borrowing behavior that favors informal credit sources. This combination challenges simplified portrayals of microfinance as an automatic driver of entrepreneurial expansion or as a complete replacement for costly informal lending.

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